Vimal Mudalagi

Cybersecurity — Security Assessment

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Education

Digital University Kerala **2025 – 2027**
Master of Science - Cybersecurity *7.9 SGPA*

K. L. E Society's G. I. Bagewadi Arts, Science and Commerce College **2022 – 2025**
Bachelor of Science - Computer Science *79.83%*

Projects

AI-Driven IoT Security Assessment and Attack Path Analysis View Project

- Building an **AI-driven IoT security assessment system** to detect network vulnerabilities, identify attack paths, and analyse potential lateral movement across emulated IoT devices.
- Emulated IoT firmware using **FirmAE** and connected devices through **Open vSwitch (OVS)**, with **OPNsense** for firewalling and Kali Linux for security testing.
- **Tech:** FirmAE, Open vSwitch, OPNsense, Kali Linux, OpenVAS

ML-Based Network Intrusion Detection Using ALNS Feature Selection View Project

- Developed a **Machine Learning (ML)-based intrusion detection system** using the CICIDS2017 dataset to classify benign traffic and cyberattacks.
- Applied **ALNS** for feature selection and trained an **MLP neural network** using ReLU and Softmax activation functions.
- Evaluated multiclass classification using **accuracy, precision, recall, F1-score, and confusion matrices**.
- **Tech:** Python, TensorFlow, Keras, Scikit-learn

Extinct Animal REST API – Dockerized Flask Service View Project

- Developed a Flask-based REST API that serves structured JSON data about extinct animals and birds, including species, extinction year, and location.
- Containerized the application with Docker and configured Gunicorn for consistent, production-style application serving.
- Implemented and tested API endpoints for retrieving animal data and integrated a browser-based interface for API exploration.
- **Tech:** Python, Flask, REST API, Docker, Gunicorn, HTML/CSS

Organizational Cyber Risk Management – Case Study View Project

- **Risk Identification**: Analyzed cybersecurity risk scenarios to identify threats, vulnerabilities, and potential business impacts.
- **Control Evaluation**: Evaluated risk management strategies and security controls to assess effectiveness in risk mitigation approaches.
- **Decision Support**: Applied risk assessment concepts to case-based security challenges, supporting risk-informed decision-making.
- **Key Concepts:** Risk Assessment, Threat Identification, Vulnerability Analysis, Risk Mitigation, Security Controls

Experience

DigiGlobe Solutions **Jun 2024 – Jul 2024**
Frontend Developer Intern (Remote)

- Developed responsive web pages and reusable UI components using React, JavaScript, HTML, and CSS.
- Collaborated with the team to implement feature updates and resolve frontend issues within project timelines.
- Improved cross-browser compatibility and usability through structured testing and UI refinements.

Digital University Kerala **Oct 2025 – Present**
Executive Member, Social Engagement Center (SEC) *Thiruvananthapuram, Kerala*

- Coordinated outreach events and student engagement activities with cross-functional teams.
- Contributed to planning and execution of community-focused initiatives, strengthening communication and teamwork.

Achievements

- **Winner – Internal Smart India Hackathon, DUK Innofest'25**
  Developed **VigilEye**, a cybersecurity solution concept for automated vulnerability assessment of CCTV cameras and DVRs.
- **Second Runner-up – Internal Smart India Hackathon, DUK Innofest'26**
  Developed **Riskyn**, a cybersecurity platform concept that quantifies cyber risk in INR, links security gaps to BPMN business processes, and prioritizes controls within a fixed budget.
- **Third Place – Boot To Rooting CTF by KSAAC, DUK'25**
  Competed in cybersecurity challenges at a Capture the Flag competition.

Certifications

CompTIA Security+ *In Progress*

Introduction to Cybersecurity – Cisco Networking Academy

Networking Basics – Cisco Networking Academy

Introduction to Python Programming for Data Science – Dataquest

Technical Skills

Mobile Application Security: Android Application Pentesting, OWASP Mobile Top 10
Security Assessment: Vulnerability Assessment, Security Testing
Threat Modeling: Threat Modeling, Attack Path Analysis
Networking: OSI Model, Network Fundamentals
Programming: Python
Databases: MySQL, MongoDB
Containerization & Orchestration: Docker, Kubernetes
Professional Skills: Problem Solving, Analytical Thinking, Team Collaboration